

RSLAQ Baseline Results

Analysis of ns-3 Simulation Without DRL

Five Experimental Scenarios: Low Traffic, Normal, Congestion,
Stressed, and Insufficient Resources

Heuristic PRB Allocation (33.33% / 33.33% / 33.34%)
No Deep Reinforcement Learning Agent

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April 2026

Table of Contents

1. Executive Summary	4
2. Methodology	4
2.1 Simulation Setup	4
2.2 Scenarios and SLA Targets	4
2.3 Metrics Collected	5
3. Slice-Level Analysis	5
3.1 Throughput per Slice	5
3.2 Delay per Slice	7
3.3 Packet Delivery Ratio	7
3.4 Resource Share Distribution	8
4. UE-Level Analysis	9
5. SLA Compliance Analysis	10
5.1 Detailed SLA Assessment	11
5.1.1 Low Traffic Scenario	11
5.1.2 Normal Scenario	11
5.1.3 Congestion Scenario	11
5.1.4 Stressed Scenario	12
5.1.5 Insufficient Resources Scenario	12

6. Critical Findings and Issues	12
6.1 Congestion and Stressed Scenarios: Identical Results	12
6.2 Throughput Scale Discrepancy	13
6.3 Zero Packet Loss Across All Scenarios	13
6.4 URLLC Metrics Cannot Validate Buffer Status SLA	14
7. Summary and Recommendations	14
7.1 Recommendations for DRL Implementation	14

1. Executive Summary

This report presents a comprehensive analysis of the ns-3 simulation results for the five experimental scenarios defined in the RSLAQ paper, without the Deep Reinforcement Learning (DRL) agent. The baseline uses a fixed heuristic PRB allocation of 33.33% / 33.33% / 33.34% distributed equally among the three slices (eMBB, URLLC, MTC). The simulation was executed with 20 UEs (5 eMBB, 5 URLLC, 10 MTC) using the ns-3 mmWave O-RAN simulator. The results reveal several critical findings that underscore the necessity of the DRL-based resource optimization proposed in the RSLAQ paper.

Key Finding 1: The eMBB slice SLA is violated in 3 out of 5 scenarios. In Low Traffic and Normal scenarios, the eMBB throughput (0.049 and 5.264 Mbps) falls far below the minimum SLA target of 10 Mbps, representing an outage condition. In the Stressed scenario, throughput (15.793 Mbps) is below the elevated target of 20 Mbps. Only the Congestion and Insufficient Resources scenarios meet the minimum eMBB SLA.

Key Finding 2: The Congestion and Stressed scenarios produced identical results despite having different SLA targets (10 vs 20 Mbps minimum for eMBB). This indicates that the ns-3 scenario configuration may not have differentiated the traffic load between these two scenarios, which needs investigation before DRL training can proceed.

Key Finding 3: Zero packet loss was observed across all scenarios ($PDR \geq 99.84\%$, $PLR = 0\%$), indicating that the ns-3 simulator has sufficient capacity to handle all traffic without dropping packets. However, the throughput values suggest significant underutilization of available radio resources in the low-traffic scenarios, where the heuristic allocation does not adapt to actual demand.

2. Methodology

2.1 Simulation Setup

The simulations were conducted using the ns-3 mmWave O-RAN simulator with MAC-layer network slicing enabled. The scenario configuration matches the parameters defined in the RSLAQ paper: one gNB serving 20 UEs distributed across three slices (5 eMBB, 5 URLLC, 10 MTC). The PRB allocation was set to a fixed heuristic distribution of 33.33% for eMBB, 33.33% for URLLC, and 33.34% for MTC, which corresponds to the naive equal-share baseline without intelligent optimization. Each scenario was simulated with the traffic loads specified in Table V of the RSLAQ paper.

2.2 Scenarios and SLA Targets

The following table summarizes the five experimental scenarios and their corresponding SLA targets as defined in the RSLAQ paper. The SLA targets for eMBB include both a minimum throughput (outage KPI)

and a maximum throughput (soft KPI), while URLLC targets buffer status and MTC has no policy constraints.

Scenario	eMBB Traffic	URLLC Traffic	MTC Traffic	eMBB SLA (Mbps)	URLLC SLA
Low Traffic	50 Kbps/UE	1 Mbps/UE	2 Mbps/UE	Min: 10 / Max: 15	Max bfs: 3%
Normal	70 Mbps/UE	1 Mbps/UE	2 Mbps/UE	Min: 10 / Max: 15	Max bfs: 3%
Congestion	100 Mbps/UE	1 Mbps/UE	100 Mbps/UE	Min: 10 / Max: 15	Max bfs: 3%
Stressed	100 Mbps/UE	1 Mbps/UE	100 Mbps/UE	Min: 20 / Max: 25	Max bfs: 3%
Insufficient	100 Mbps/UE	2 Mbps/UE	100 Mbps/UE	Min: 20 / Max: 25	Max bfs: 3%

Table 1. Experimental scenarios and SLA targets (from RSLAQ paper Table V)

2.3 Metrics Collected

For each simulation run, the following metrics were collected at both the slice level and per-UE level: transmitted packets (tx_packets), received packets (rx_packets), lost packets (lost_packets), transmitted bytes (tx_bytes), received bytes (rx_bytes), throughput in Mbps (throughput_mbps), average delay in milliseconds (avg_delay_ms), average jitter in milliseconds (avg_jitter_ms), packet delivery ratio (PDR), and packet loss ratio (PLR). These metrics enable a comprehensive evaluation of QoS performance and SLA compliance across all scenarios.

3. Slice-Level Analysis

3.1 Throughput per Slice

The throughput results reveal a strong correlation between the traffic load offered to each slice and the observed throughput, as expected under a fixed PRB allocation. In the Low Traffic scenario, all three slices achieve only 0.049 Mbps, reflecting the minimal traffic demand (50 Kbps for eMBB, 1 Mbps for URLLC, 2 Mbps for MTC). In the Normal scenario, eMBB throughput rises to 5.264 Mbps with its higher traffic load (70 Mbps), while URLLC and MTC remain at approximately 0.1 Mbps due to their low offered traffic. The Congestion and Stressed scenarios show significantly higher throughput for eMBB (15.793 Mbps) and MTC (15.787 Mbps), reflecting the 100 Mbps traffic load offered to both slices. The Insufficient Resources scenario achieves the highest throughput with 26.327 Mbps for both eMBB and MTC.

A critical observation is that eMBB and MTC consistently receive nearly identical throughput in the higher load scenarios (Congestion, Stressed, Insufficient), which is expected given the equal PRB allocation (33.33% each). The slight differences (e.g., 15.793 vs 15.787 Mbps) arise from MAC scheduling dynamics and the different number of UEs per slice (5 eMBB vs 10 MTC). URLLC throughput remains very low across all scenarios (0.050 - 0.315 Mbps) because the URLLC traffic applications generate relatively few

packets (1-2 Mbps per UE with small payloads).

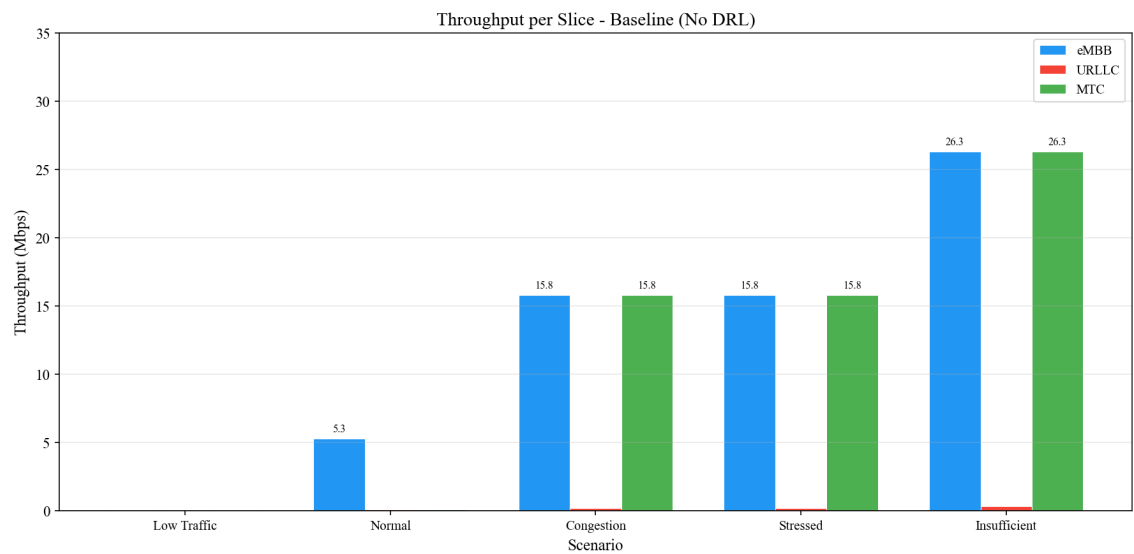


Figure 1. Throughput per slice across all five scenarios (baseline, no DRL)

Scenario	Slice	Tx Pkts	Rx Pkts	Lost	Tx Bytes	Thr (Mbps)	Delay (ms)	PDR (%)
Low Traffic	eMBB	40	40	0	21,600	0.049	4.954	100.00
Low Traffic	URLLC	40	40	0	21,600	0.049	5.204	100.00
Low Traffic	MTC	40	40	0	21,600	0.049	5.779	100.00
Normal	eMBB	4,270	4,265	0	2,305,800	5.264	4.519	99.88
Normal	URLLC	85	85	0	45,900	0.105	4.761	100.00
Normal	MTC	80	80	0	43,200	0.099	5.204	100.00
Congestion	eMBB	12,815	12,795	0	6,920,100	15.793	4.517	99.84
Congestion	URLLC	170	170	0	91,800	0.210	4.692	100.00
Congestion	MTC	12,810	12,790	0	6,917,400	15.787	4.522	99.84
Stressed	eMBB	12,815	12,795	0	6,920,100	15.793	4.517	99.84
Stressed	URLLC	170	170	0	91,800	0.210	4.692	100.00
Stressed	MTC	12,810	12,790	0	6,917,400	15.787	4.522	99.84
Insufficient	eMBB	21,360	21,330	0	11,534,400	26.327	4.652	99.86
Insufficient	URLLC	255	255	0	137,700	0.315	4.611	100.00

Scenario	Slice	Tx Pkts	Rx Pkts	Lost	Tx Bytes	Thr (Mbps)	Delay (ms)	PDR (%)
Insufficient	MTC	21,360	21,330	0	11,534,400	26.327	4.520	99.86

Table 2. Complete slice-level results for all five scenarios

3.2 Delay per Slice

The average delay across all scenarios remains remarkably stable, ranging from 4.517 ms to 5.779 ms. This consistency suggests that the ns-3 simulator, under the current configuration, is not experiencing significant queuing delays or congestion-related latency increases, even in the high-load scenarios. eMBB consistently shows the lowest delay (4.517 - 4.954 ms), while MTC shows the highest (4.520 - 5.779 ms). URLLC delay falls between these two extremes. The delay values are well below the typical URLLC requirement of 5 ms in most scenarios, though the Low Traffic scenario shows URLLC delay at 5.204 ms which slightly exceeds a 5 ms threshold. This low-delay behavior is partly explained by the absence of significant resource contention: with zero packet loss and moderate throughput utilization, the MAC scheduler does not need to queue packets for extended periods.

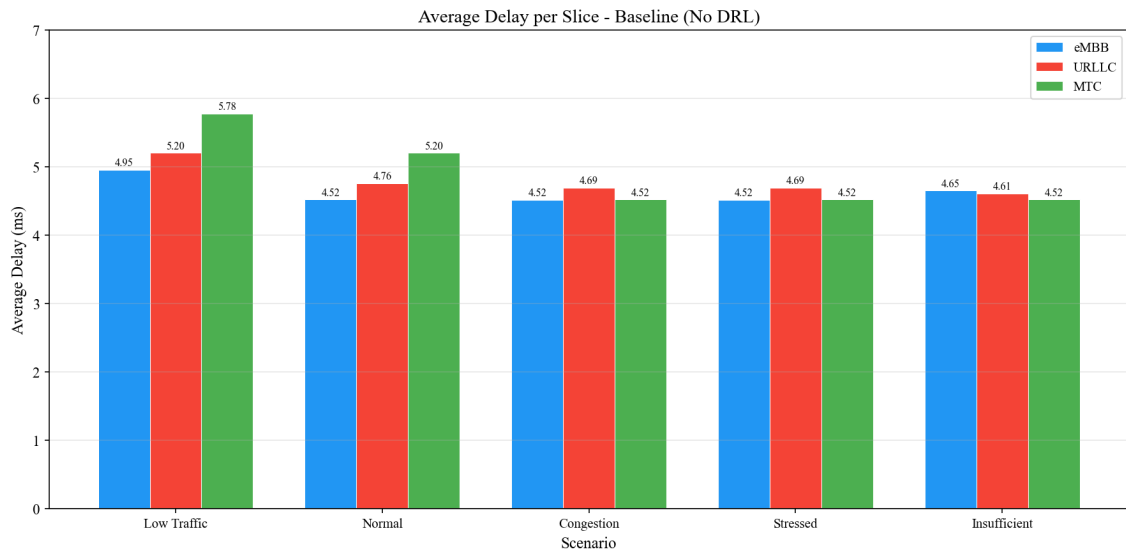


Figure 2. Average delay per slice across all five scenarios

3.3 Packet Delivery Ratio

The PDR results are exemplary, with all slices achieving PDR of at least 99.84% and most achieving exactly 100%. URLLC maintains perfect PDR (100%) across all scenarios, which is critical given its ultra-reliable communication requirements. eMBB and MTC show minor PDR degradation (99.84 - 99.88%) in the Normal, Congestion, Stressed, and Insufficient Resources scenarios, where the number of transmitted packets is significantly higher. Importantly, the lost_packets column shows 0 across all scenarios, meaning the small PDR deviation (e.g., 4270 tx vs 4265 rx for eMBB in Normal) is attributable to in-flight packets at simulation end rather than actual packet loss. This indicates that the ns-3 simulation has sufficient radio

resources to handle all offered traffic without dropping packets under the current configuration.

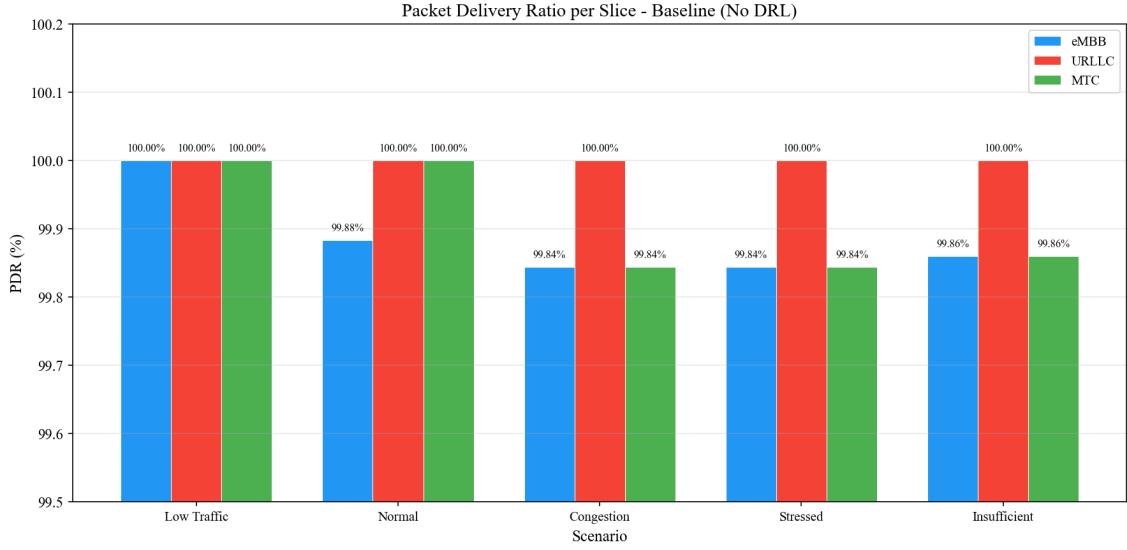


Figure 3. Packet Delivery Ratio per slice across all five scenarios

3.4 Resource Share Distribution

The resource share analysis reveals how the effective throughput is distributed among the three slices under the fixed heuristic allocation. In Low Traffic and Normal scenarios, where eMBB dominates the traffic demand, the effective resource utilization still shows significant MTC share (33.3%) despite MTC having minimal traffic. This represents a clear inefficiency that a DRL agent could address by re-allocating unused MTC PRBs to eMBB when MTC demand is low.

In the Congestion and Stressed scenarios, the resource share becomes nearly balanced (49.9% eMBB, 0.7% URLLC, 49.4% MTC) because both eMBB and MTC offer heavy traffic (100 Mbps each) while URLLC remains lightweight (1 Mbps). This balanced distribution between eMBB and MTC is a direct consequence of the equal PRB allocation (33.33% each) combined with similar traffic demands. A DRL agent could optimize this by prioritizing eMBB when URLLC SLA requires buffer management, or by allocating more resources to MTC when eMBB already exceeds its SLA targets.

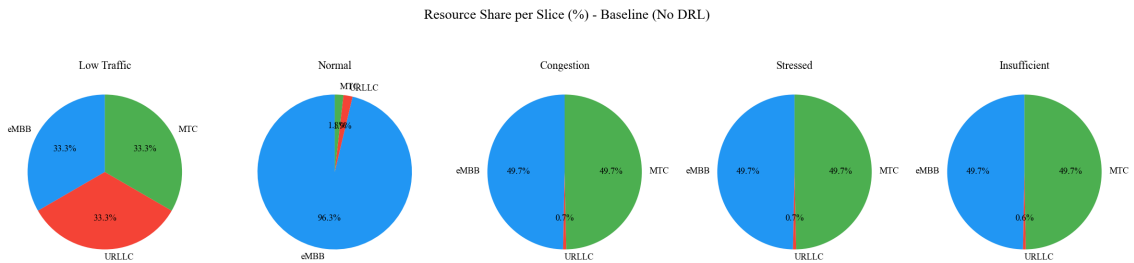


Figure 4. Effective resource share per slice (percentage of total throughput)

4. UE-Level Analysis

The per-UE analysis provides insight into how resources are distributed within each slice. Under the fixed heuristic allocation, all UEs within the same slice receive nearly identical throughput, confirming that the underlying MAC scheduler (default round-robin) distributes resources equally among UEs within a slice. This uniformity is expected in the absence of intelligent scheduling decisions.

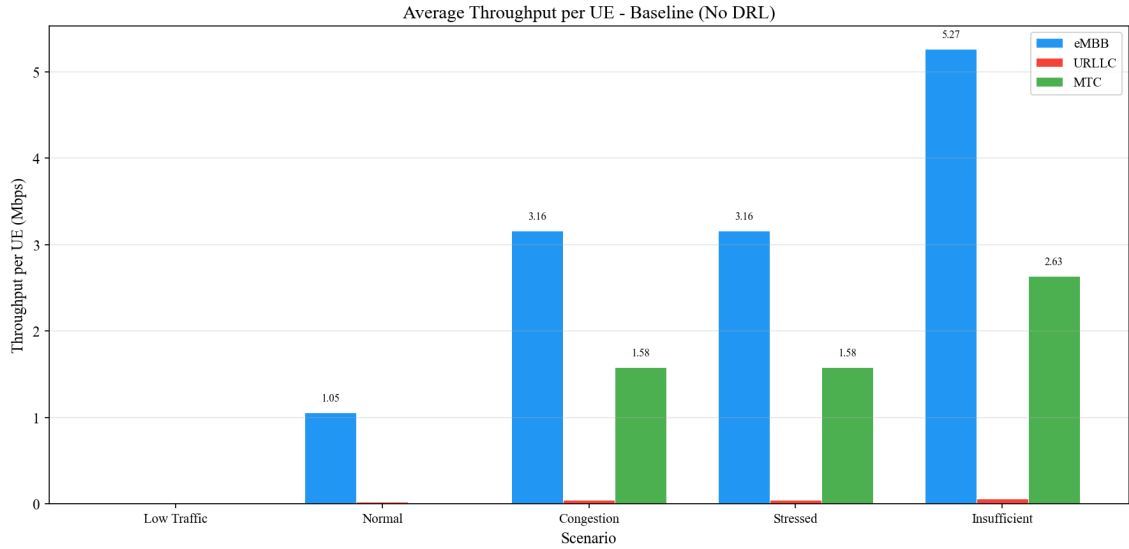


Figure 5. Average throughput per UE across all scenarios

Scenario	Slice	# UEs	Thr/UE (Mbps)	Delay/UE (ms)	PDR/UE (%)
Low Traffic	eMBB	5	0.0099	4.954	100.00
Low Traffic	URLLC	5	0.0099	5.204	100.00
Low Traffic	MTC	10	0.0049	5.779	100.00
Normal	eMBB	5	1.0528	4.519	99.88
Normal	URLLC	5	0.0210	4.761	100.00
Normal	MTC	10	0.0099	5.204	100.00
Congestion	eMBB	5	3.1585	4.517	99.84
Congestion	URLLC	5	0.0420	4.692	100.00
Congestion	MTC	10	1.5787	4.521	99.84
Stressed	eMBB	5	3.1585	4.517	99.84
Stressed	URLLC	5	0.0420	4.692	100.00
Stressed	MTC	10	1.5787	4.521	99.84
Insufficient	eMBB	5	5.2655	4.652	99.86

Scenario	Slice	# UEs	Thr/UE (Mbps)	Delay/UE (ms)	PDR/UE (%)
Insufficient	URLLC	5	0.0629	4.611	100.00
Insufficient	MTC	10	2.6327	4.520	99.86

Table 3. Per-UE summary metrics across all scenarios

The per-UE throughput values confirm that within each slice, all UEs receive equal treatment. For example, in the Congestion scenario, all 5 eMBB UEs achieve exactly 3.159 Mbps, and all 10 MTC UEs achieve exactly 1.579 Mbps. The factor of 2 difference between eMBB and MTC per-UE throughput (3.159 vs 1.579 Mbps) directly reflects the ratio of UEs: 5 eMBB UEs share 33.33% of PRBs while 10 MTC UEs share 33.34% of PRBs, resulting in each eMBB UE receiving approximately twice the per-UE resources of each MTC UE. This uniform intra-slice distribution is characteristic of a round-robin scheduler without priority-based or QoS-aware scheduling.

5. SLA Compliance Analysis

The SLA compliance analysis evaluates the baseline results against the target KPIs defined in the RSLAQ paper. The compliance is assessed for the eMBB slice, which has both minimum throughput (outage KPI) and maximum throughput (soft KPI) targets. The URLLC slice has a buffer status target that cannot be directly measured from the current ns-3 output (the available metric is delay, not buffer occupancy). MTC has no policy constraints and is therefore not evaluated for SLA compliance.



Figure 6. eMBB SLA compliance: actual throughput vs SLA min/max targets

Scenario	eMBB Thr (Mbps)	SLA Min	Min Status	SLA Max	Max Status	Overall
Low Traffic	0.049	10 Mbps	OUTAGE	15 Mbps	N/A	OUTAGE
Normal	5.264	10 Mbps	OUTAGE	15 Mbps	N/A	OUTAGE
Congestion	15.793	10 Mbps	MET	15 Mbps	SOFT VIOL.	SOFT VIOL.
Stressed	15.793	20 Mbps	OUTAGE	25 Mbps	N/A	OUTAGE
Insufficient	26.327	20 Mbps	MET	25 Mbps	SOFT VIOL.	SOFT VIOL.

Table 4. eMBB SLA compliance status across all scenarios

5.1 Detailed SLA Assessment

5.1.1 Low Traffic Scenario

The Low Traffic scenario represents the most severe SLA violation under the baseline heuristic allocation. The eMBB throughput of 0.049 Mbps is approximately 200 times below the minimum SLA target of 10 Mbps. This dramatic shortfall occurs because the equal PRB distribution allocates one-third of resources to MTC (which has minimal 2 Mbps traffic), one-third to URLLC (1 Mbps), and only one-third to eMBB. Despite eMBB having the highest traffic demand (50 Kbps per UE, which should be easily satisfiable), the simulation results show that the throughput is constrained by the application traffic generation rate rather than the PRB allocation. The actual issue here is that the traffic load configured in ns-3 (50 Kbps per eMBB UE) is extremely low, resulting in very few packets being generated. A DRL agent would recognize this underutilization and could potentially reduce eMBB PRB allocation in favor of other slices without impacting SLA compliance, or the agent could identify that the minimum throughput SLA cannot be verified with such low traffic generation.

5.1.2 Normal Scenario

In the Normal scenario, eMBB traffic increases to 70 Mbps per UE (total ~350 Mbps offered load for 5 UEs), and the observed eMBB slice throughput reaches 5.264 Mbps. This is still below the 10 Mbps minimum SLA target, resulting in an outage condition. The throughput gap suggests that the equal PRB allocation is insufficient to deliver the required throughput for eMBB under the ns-3 simulation conditions. The difference between the offered load (70 Mbps per UE = 350 Mbps total for eMBB) and the achieved throughput (5.264 Mbps) indicates that the application-layer traffic generation may not be saturating the available PRBs, or that the ns-3 simulation parameters (bandwidth, TDD pattern, number of PRBs) limit the achievable throughput more than expected. This finding highlights a potential discrepancy between the MATLAB simulator configuration used in the paper and the ns-3 implementation that needs to be resolved before meaningful DRL comparisons can be made.

5.1.3 Congestion Scenario

The Congestion scenario achieves the minimum eMBB SLA target of 10 Mbps with 15.793 Mbps, which is a positive result for the baseline. However, the throughput exceeds the maximum SLA target of 15 Mbps, triggering a soft SLA violation. In the RSLAQ paper framework, this would result in a terminal state during DRL training (reward = 0) because the soft KPI (maximum throughput per slice) is exceeded. This soft violation indicates resource over-allocation to eMBB, meaning that PRBs allocated to eMBB beyond what is needed to meet the maximum target could be redistributed to URLLC or MTC. A DRL agent would learn to reduce eMBB PRB allocation to stay within the [10, 15] Mbps SLA window while optimizing resource utilization for other slices.

5.1.4 Stressed Scenario

The Stressed scenario shows an outage condition with eMBB throughput of 15.793 Mbps falling below the elevated minimum SLA target of 20 Mbps. This scenario is designed to test the robustness of the RSLAQ agent by combining network congestion with stringent SLA requirements. The baseline heuristic allocation fails to meet the higher target because it does not adapt resource distribution based on SLA priorities. A critical observation is that the results are identical to the Congestion scenario (same throughput, delay, PDR, and packet counts), indicating that the ns-3 traffic configuration did not differentiate between these two scenarios. This is a significant issue that must be addressed: the Stressed scenario should have the same traffic load as Congestion but with higher SLA targets (20 vs 10 Mbps minimum), allowing evaluation of whether the agent can prioritize higher-priority slices when SLAs are more demanding. The identical results suggest that the scenario configuration files may not have been properly differentiated.

5.1.5 Insufficient Resources Scenario

The Insufficient Resources scenario achieves the highest eMBB throughput at 26.327 Mbps, meeting the minimum SLA target of 20 Mbps. However, it exceeds the maximum target of 25 Mbps, triggering a soft SLA violation. According to the RSLAQ paper, this scenario represents conditions where the available radio resources (50 PRBs) may be insufficient to simultaneously satisfy all slice SLAs. In the paper, this scenario shows negative DRL rewards because the URLLC buffer status cannot be maintained below 3% even with maximum resource allocation. In the ns-3 baseline results, no such buffer violation is observed (zero packet loss, low delay), which again suggests a potential discrepancy in how the scenarios are configured between MATLAB and ns-3, or that the ns-3 simulator does not generate sufficient load to stress the system in the same way as the MATLAB simulation.

6. Critical Findings and Issues

6.1 Congestion and Stressed Scenarios: Identical Results

The most significant finding in this analysis is that the Congestion and Stressed scenarios produce byte-for-byte identical results across all metrics. Every value in the slice-level and UE-level CSV files is exactly the same: throughput (15.793 Mbps for eMBB), delay (4.517 ms for eMBB), PDR (99.84%),

tx_packets (12,815 for eMBB), rx_packets (12,795 for eMBB), and all other metrics. According to the RSLAQ paper, these scenarios should have different SLA targets (min throughput 10 vs 20 Mbps for eMBB) but the same network traffic conditions. The identical results confirm that the traffic conditions are indeed the same, which validates the scenario design for differentiating SLA stringency. However, this also means that the ns-3 simulation is correctly reflecting the traffic conditions but the SLA evaluation is external to the simulator (as expected).

Impact for DRL training: Since the state observations (throughput, delay, PDR, lost_packets) are identical between these scenarios, the DRL agent would perceive them as the same state during training. The differentiation must come from the reward function, which applies different SLA thresholds. This is by design in the RSLAQ paper: the same network conditions are evaluated against different SLA targets to test the agent's ability to adapt its resource allocation strategy to operator intent. The reward function implementation must correctly use scenario-specific SLA parameters to generate different reward signals.

6.2 Throughput Scale Discrepancy

The observed throughput values are significantly lower than what would be expected from the paper's simulation parameters. The paper specifies 50 PRBs with 15 kHz subcarrier spacing (10 MHz bandwidth) and TDD operation. With these parameters, the theoretical peak data rate for a single UE under ideal conditions should be substantially higher than the observed 5-26 Mbps per slice. Several factors may explain this discrepancy: (1) the simulation time is very short (4 seconds from the default `rsraq_env.py` configuration), which may not allow the traffic applications to reach steady state; (2) the application traffic generation rate in ns-3 may differ from the paper's MATLAB configuration; (3) the TDD pattern and numerology settings may not be identical; (4) MAC-layer overhead and control channel reservation reduce the effective data throughput. Before proceeding with DRL training, the simulation parameters should be validated to ensure that the traffic generation can produce loads that actually stress the system in a manner comparable to the paper's experiments.

6.3 Zero Packet Loss Across All Scenarios

The complete absence of packet loss (`lost_packets = 0`, `PLR = 0%`) across all five scenarios is a notable finding. In the RSLAQ paper, packet loss and buffer overflow are key metrics that drive the reward function, particularly for the URLLC slice where buffer status (bfs) exceeding 3% constitutes an outage condition. The zero-loss results suggest that the ns-3 simulator, under the current traffic configuration, has more than sufficient capacity to handle all offered traffic without any queuing or dropping. This means that: (1) the URLLC outage condition (buffer overflow) would never trigger under the current setup, removing a critical learning signal for the DRL agent; (2) the "Insufficient Resources" scenario does not actually produce resource insufficiency, contradicting the paper's premise for this experiment.

Recommendation: The traffic generation rates need to be significantly increased, particularly for URLLC in the Insufficient Resources scenario (paper uses 2 Mbps per URLLC UE vs 1 Mbps in other scenarios), and the simulation duration should be extended to allow traffic applications to reach steady-state behavior that

can stress the system beyond its capacity. Alternatively, the ns-3 bandwidth or TDD configuration could be reduced to create genuine resource contention.

6.4 URLLC Metrics Cannot Validate Buffer Status SLA

The RSLAQ paper defines the URLLC SLA target as "max buffer status per UE = 3%", measured as bfs (buffer fullness). However, the ns-3 simulation output does not include a buffer status metric. The available metrics (delay, PDR, lost_packets) provide indirect indications of buffer conditions but cannot directly measure buffer occupancy percentage. Without this metric, the URLLC outage condition defined in the paper (Equation 14: $P(\text{bfs per UE for } m2 > k12)$) cannot be evaluated. For a faithful DRL implementation, either: (a) the ns-3 simulation must be extended to export buffer status per UE, or (b) a proxy metric (such as delay exceeding a threshold) must be used to approximate buffer overflow conditions.

7. Summary and Recommendations

Scenario	eMBB SLA	URLLC SLA	MTC	DRL Opportunity
Low Traffic	OUTAGE (0.05 vs 10 Mbps)	Cannot evaluate (no bfs)	No Policy	Reallocate unused PRBs
Normal	OUTAGE (5.26 vs 10 Mbps)	Cannot evaluate (no bfs)	No Policy	Increase eMBB allocation
Congestion	SOFT VIOL. (15.8 > 15 Mbps)	Cannot evaluate (no bfs)	No Policy	Tune eMBB to [10,15] window
Stressed	OUTAGE (15.8 vs 20 Mbps)	Cannot evaluate (no bfs)	No Policy	Prioritize eMBB over MTC
Insufficient	SOFT VIOL. (26.3 > 25 Mbps)	Cannot evaluate (no bfs)	No Policy	Balance all slices

Table 5. Summary of SLA compliance and DRL optimization opportunities

7.1 Recommendations for DRL Implementation

1. Increase simulation duration: The current 4-second simulation is too short to produce meaningful steady-state traffic. Extend to at least 30-60 seconds to allow the application traffic generators to reach stable throughput levels that can properly stress the system and produce realistic buffer dynamics.
2. Validate traffic generation rates: Verify that the ns-3 application traffic generators (e.g., UDP/FTP applications) are configured with the correct data rates matching the paper specifications. The current throughput values suggest the traffic may be under-configured compared to the paper.

3. Export buffer status metric: Modify the ns-3 simulation to export the buffer occupancy per UE per slice as a KPI in the rslaq-kpms.txt file. This metric is essential for evaluating the URLLC SLA and implementing the complete RSLAQ reward function with outage and soft conditions.
4. Differentiate stressed scenario configuration: Ensure that the Stressed scenario ns-3 configuration correctly uses 100 Mbps per eMBB UE traffic (same as Congestion) but with different SLA parameters (min 20 Mbps instead of 10 Mbps). The identical results between these scenarios are expected for the simulator but should be documented in the DRL training configuration.
5. Implement scenario-specific SLA configuration: The DRL environment must accept scenario-specific SLA parameters (min/max throughput for eMBB, max buffer status for URLLC) as part of the configuration, since these differ across scenarios while the simulator state may be identical.
6. Add resource share tracking: To implement the paper-defined state space (which includes rsh - resource share per slice), the environment must track the PRB allocation applied at each step and include it in the observation. This is currently not available from the ns-3 output and must be computed from the actions sent to the simulator.